Fed-ZoomSIB with a server: what is (and is not) new

Proofs for claims 1.2, 1.3 and 2.1 of `docs/claims-evidence.md`

CS6007 project notes

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Abstract

We make precise the statement "with a server, N agents for T rounds are a single agent for NT rounds". Theorem 3 shows that no N-agent algorithm, whatever it communicates, can beat the single-agent minimax rate at horizon NT. Theorem 9 shows that Fed-ZoomSIB with exact sufficient-statistic pooling and event-triggered sync *attains* that rate. Theorem 12 bounds its communication by $O(N_{\mathrm{b}} \log_{1+\gamma}(NT))$ sync rounds, and Proposition 15 shows that Phase 1 is an $O(d/(NT)^{1/3})$ fraction of all traffic. Section 5 sketches the serverless (relay) version, whose bound carries an additive $O(NDN_{\mathrm{b}})$ term, and Section 6 states the open regret–communication question these results leave. Imported results from Dey, Bhore & Ghosh (2026), hereafter *DBG*, are marked as such.

1 Setting

N agents, T rounds. At round t agent i observes K contexts $x_{i,t,1}, \dots, x_{i,t,K}$ drawn i.i.d. from $\mathcal{D}$ on $\mathbb{R}^d$, independently across agents and rounds, pulls $a_{i,t}$ and observes $y_{i,t} = f(\langle x_{i,t,a_{i,t}}, \theta^* \rangle) + \eta_{i,t}$. Both θ^* (with $\|\theta^*\|_1 = 1$) and f are unknown and shared by all agents. Network regret is

$$R_T(N) = \sum_{i=1}^{N} \sum_{t=1}^{T} \Big[\max_a f(\langle x_{i,t,a}, \theta^* \rangle) - f(\langle x_{i,t,a_{i,t}}, \theta^* \rangle) \Big].$$

Assumption 1 (DBG Assumptions 1–3)**.** (i) $\eta_{i,t}$ is independent of the contexts and conditionally σ-sub-Gaussian given the past. (ii) $\mathbb{E}[S_j(X)^2] \le M$ for the score $S = -\nabla \log p$. (iii) Each coordinate of X is κ-sub-Gaussian; $|f| \le L_f$ and $|f'| \le L_{f'}$ on $|z| \le X_{\max}$, where $X_{\max} = \kappa\sqrt{2\log(2dNTK/\delta)}$ bounds $\|x\|_\infty$ for all NTK contexts with probability $\ge 1 - \delta$. (iv) $\mu^* = \mathbb{E}[f'(\langle X, \theta^* \rangle)] > 0$.

Lemma 1 (Imported: DBG Lemma B.1 with Lemma 2.1)**.** *Let $\hat\theta$ be the truncated Stein estimator $\frac{1}{n}\sum_{s=1}^{n} \varphi_\tau(y_s S(x_s))$ on n i.i.d. pairs with $\tau = \sqrt{3(\sigma^2 + L_f^2) M n / \log(2d/\delta)}$, and $\hat\theta_0 = \hat\theta / \|\hat\theta\|_1$. If n is large enough that $C_\theta d\sqrt{\log(2d/\delta)/n} \le \mu^*/2$, then with probability $\ge 1 - \delta$*

$$\|\hat\theta_0 - \theta^*\|_1 \le \varepsilon_0(n) := \frac{4 C_\theta d}{\mu^*} \sqrt{\frac{\log(2d/\delta)}{n}}, \qquad C_\theta = \tfrac{4\sqrt{3}}{3} + 2\sqrt{2}\sqrt{M(\sigma^2 + L_f^2)}.$$

Lemma 2 (Imported: DBG Theorem 5.1)**.** *There are $c > 0$ and n_0 such that for every horizon $n \ge n_0$ and every single-agent policy π, $\sup \mathbb{E}[R_n(\pi)] \ge c\, n^{2/3}$, the supremum over instances satisfying the regularity assumptions (with $d = 1$, Gaussian contexts and $K = \lceil 4N' \log n / p_{\min} \rceil$ arms for an N'-bin construction).*

2 Lower bound: a network is never better than one agent with all the data

Theorem 3 (Lower bound by simulation)**.** *Let $\mathcal{A}$ be any N-agent algorithm, with arbitrary communication, including every agent seeing every other agent's data after every round. Then for $NT \geq n_0$,*

$$\sup_{instances} \mathbb{E}[R_T(N;\mathcal{A})] \;\geq\; c\,(NT)^{2/3}.$$

The same holds for every serverless algorithm, since it is a special case.

Proof. We build a single-agent policy π with horizon $n = NT$ whose regret has exactly the law of $\mathcal{A}$'s network regret; Lemma 2 then applies to π.

Index the single agent's rounds by $u = (t-1)N + i$, so that round u plays the role of "agent i at round t". Policy π keeps a copy of $\mathcal{A}$'s full internal state (all agents, the server, and their random seeds, which π draws itself). At round $u = (t, i)$:

1. π observes its context set X_u and passes it to the simulated agent i, which selects a_u using its state *as of the end of round* $t-1$, exactly as in $\mathcal{A}$.
2. π plays a_u and receives y_u, but *withholds* it from the simulation.
3. When $i = N$, all N outcomes of round t are available. π feeds them to the simulated agents' updates and runs one round of $\mathcal{A}$'s communication.

This is a valid policy: each decision uses only information from strictly earlier rounds of π.

The context sets X_u are i.i.d. from $\mathcal{D}^K$, and the noise is independent of everything before it, in both systems. So by induction over t, the joint law of (contexts, actions, rewards) of round t is the same under π and under $\mathcal{A}$, given the same law up to round $t-1$. In particular $R_{NT}(\pi) \overset{d}{=} R_T(N;\mathcal{A})$ on every instance. Hence $\sup \mathbb{E}R_T(N;\mathcal{A}) = \sup \mathbb{E}R_{NT}(\pi) \geq c(NT)^{2/3}$. □

Remark 4. The argument uses only that agents are statistically identical (shared $\theta^*, f, \mathcal{D}$) and act in parallel. It fails under concept shift (different θ_i^*), which is where collaboration stops being "one agent with more data".

3 Upper bound: Fed-ZoomSIB-SS attains the single-agent rate at NT

Algorithm (as analysed).

- *Phase 1* (rounds $1..T_0$): every agent pulls uniformly at random and stores $(S(x), y)$. At round T_0 each agent uploads $b_i = \sum_{t \leq T_0} \varphi_\tau(y_{i,t} S(x_{i,t}))$ and $n_i = T_0$, with τ set for the pooled size $n = NT_0$. The server broadcasts $\hat{\theta}_0 = \sum_i b_i / \|\sum_i b_i\|_1$. (Code: `phase1="exact"`, `freeze="fixed"`.)
- *Grid*: window $W = X_{\max}$; bins $I_j = (-W + (j-1)\Delta,\, -W + j\Delta]$ for $j \leq N_{\mathrm{b}} = \lceil 2W/\Delta \rceil$. An arm x belongs to bin $b(x) \ni \langle x, \hat{\theta}_0 \rangle$.
- *Phase 2*: the server holds the synced table (G_j, G_j^S). Agent i holds unsynced deltas $(\delta n_{i,j}, \delta S_{i,j})$ and acts on $n_{i,j} = G_j + \delta n_{i,j}$, $S_{i,j} = G_j^S + \delta S_{i,j}$. Among the bins its arms fall into, it picks $\arg\max_j \, S_{i,j}/n_{i,j} + w(n_{i,j})$ (with $w(0) = \infty$) and any arm in that bin.
- *Trigger*: at the end of a round, if some agent has $\delta n_{l,j} > \gamma \max(G_j, 1)$, all agents upload their deltas and the server merges and broadcasts them. (Code: `phase2="event"`.)

The confidence width is $w(n) = c_\delta \sqrt{2/n}$ with $c_\delta = \sigma\sqrt{2\log(4NN_{\mathrm{b}}(NT)^3/\delta)}$. (The code uses the smaller $\sigma\sqrt{2\log(2N_{\mathrm{b}}NT/\delta)}$; the difference is a constant and a log factor.)

Lemma 5 (Phase 1 is exactly centralised)**.** $\sum_i b_i/(NT_0)$ *is the truncated Stein estimator on the pooled* $n = NT_0$ *samples. These are i.i.d. with the law of single-agent Phase-1 samples. Hence, on an event* E_θ *of probability* $\geq 1-\delta$, $\|\hat{\theta}_0 - \theta^*\|_1 \leq \varepsilon_0(NT_0)$.

Proof. Truncation is applied per summand with the same τ, so the sum of agents' sums is the sum over the pooled sample. Uniform pulls make each (x, y) an independent draw of the context distribution and the reward model. Apply Lemma 1. □

Good events. E_x: $\|x\|_\infty \leq X_{\max}$ for all contexts. E_θ: as in Lemma 5. E_η: the noise concentration defined in Lemma 6. On $E_x \cap E_\theta$,

$$|\langle x, \hat{\theta}_0\rangle - \langle x, \theta^*\rangle| \leq \|x\|_\infty \|\hat{\theta}_0 - \theta^*\|_1 \leq X_{\max}\varepsilon_0 =: \varepsilon_{\mathrm{p}}, \tag{1}$$

and every arm lies in the window, since $|\langle x, \hat{\theta}_0\rangle| \leq \|x\|_\infty \|\hat{\theta}_0\|_1 \leq W$. Let $\tilde{I}_j = I_j + [-\varepsilon_{\mathrm{p}}, \varepsilon_{\mathrm{p}}]$ and $m_j = \sup_{z\in\tilde{I}_j} f(z)$. By (1) every pull into bin j has true index in $\tilde{I}_j$, so its mean $\mu_s = f(z_s)$ lies in $[m_j - \omega, m_j]$, where $\omega = L_{f'}(\Delta + 2\varepsilon_{\mathrm{p}})$.

Lemma 6 (Confidence for stale, agent-specific tables)**.** *Order all Phase-2 pulls into bin* j *lexicographically by (round, agent). With probability* $\geq 1-\delta$ *(event* E_η*), for every agent* i*, bin* j *and round* t*,*

$$\left|\tfrac{S_{i,j}}{n_{i,j}} - \tfrac{1}{n_{i,j}}\textstyle\sum_{s\in seen}\mu_s\right| \leq w(n_{i,j}).$$

Proof. The pulls agent i sees in bin j are a *prefix* of bin j's sequence (everything up to the last sync) together with a *contiguous segment* of agent i's own bin-j pulls (since that sync). The noise along the lexicographic order is a martingale difference sequence: the bin chosen at (t, i) depends only on rewards of earlier rounds. Azuma–Hoeffding with a union bound over all prefixes ($\leq NT$ per bin) and all segments ($\leq (NT)^2$ per agent and bin) bounds each part's noise sum by $c_\delta\sqrt{\text{length}}$. Then $\sqrt{p} + \sqrt{q} \leq \sqrt{2(p+q)}$ completes the proof. □

Lemma 7 (Per-pull regret)**.** *On* $E = E_x \cap E_\theta \cap E_\eta$*, if agent* i *pulls from bin* J *at round* t*, its instant regret is at most* $\min\{2L_f,\ 2w(n_{i,J}) + 2\omega\}$*. This holds for* any *availability pattern.*

Proof. The best arm x^* lies in an available bin j^* with $f(\langle x^*, \theta^*\rangle) \leq m_{j^*}$. By Lemma 6,

$$\mathrm{UCB}_{j^*} \geq (m_{j^*} - \omega) - w + w = m_{j^*} - \omega,$$

or $\mathrm{UCB}_{j^*} = \infty$. Also $\mathrm{UCB}_J \leq m_J + 2w(n_{i,J})$. The pulled arm has mean $\geq m_J - \omega$. Combining with $\mathrm{UCB}_J \geq \mathrm{UCB}_{j^*}$ gives regret

$$\leq (\mathrm{UCB}_J + \omega) - (m_J - \omega) \leq 2w(n_{i,J}) + 2\omega.$$

Rewards are in $[-L_f, L_f]$, which gives the other bound. □

Lemma 8 (Stale counts)**.** *If a pull is the* k*-th pull into bin* J *(lexicographic order), then the count its agent acts on satisfies* $n_{i,J} \geq (k - c_N)/(1 + N\gamma)$*, where* $c_N = N(1+\gamma)$.

Proof. The $k-1$ earlier pulls are of three kinds:

- synced, G_J;
- agent i's own unsynced pulls;
- other agents' pulls agent i has not seen.

The last kind are either unsynced, at most $(N-1)\gamma\max(G_J, 1)$ because no trigger fired at the end of round $t-1$, or from round t itself, at most $N-1$. Since $n_{i,J} \geq G_J$:

- if $G_J \geq 1$, then $k - 1 \leq n_{i,J}(1 + N\gamma) + N$;
- if $G_J = 0$, then $n_{i,J} \geq k - 1 - (N - 1)(1 + \gamma)$.

Both cases imply the claim. □

Theorem 9 (Upper bound)**.** *Under Assumption 1, with $\delta = 1/(NT)$, $\Delta = (NT)^{-1/3}$, any trigger threshold $\gamma > 0$, and $NT_0 = \lceil (L_{f'} X_{\max} \frac{4C_\theta d}{\mu^*} \sqrt{\log(2dNT)}\, NT/L_f)^{2/3} \rceil$ (and at least the size Lemma 1 requires),*

$$\mathbb{E}R_T(N) \leq \underbrace{2L_f NT_0}_{\text{Phase 1}} + \underbrace{4\sqrt{2}\, c_\delta \sqrt{(1 + N\gamma)N_{\mathrm{b}} NT}}_{\text{UCB}} + \underbrace{2L_{f'} \Delta NT}_{\text{discretisation}} + \underbrace{4L_{f'} X_{\max} \varepsilon_0(NT_0) NT}_{\text{projection}} + \underbrace{2L_f N_{\mathrm{b}}(c_N + 1)}_{\text{staleness}} + 6L_f,$$

With $\gamma = \gamma_0/N$ for a constant γ_0 this is $\tilde{O}\big((d\,NT)^{2/3} + N(NT)^{1/3}\big) = \tilde{O}\big(d^{2/3}(NT)^{2/3}\big)$ for $N \leq \sqrt{T}$. This matches Theorem 3 in N and T up to logarithms: the server setting is "one agent with NT pulls", and no faster.

Proof. Phase-1 pulls cost at most $2L_f$ each. On the failure event (probability $\leq 3\delta$) regret is at most $2L_f NT$, giving the $6L_f$ term. On E, sum Lemma 7 over the pulls into bin j, using Lemma 8 for the k-th pull:

$$\sum_{k=1}^{n_j} \min\left\{2L_f,\ 2c_\delta \sqrt{\tfrac{2(1+N\gamma)}{k - c_N}} + 2\omega\right\} \leq 2L_f(c_N + 1) + 4\sqrt{2}\, c_\delta \sqrt{(1 + N\gamma)n_j} + 2\omega n_j.$$

Summing over bins, Cauchy–Schwarz gives $\sum_j \sqrt{n_j} \leq \sqrt{N_{\mathrm{b}} NT}$. Expanding ω gives the discretisation and projection terms. With $\Delta = (NT)^{-1/3}$ we have $N_{\mathrm{b}} \leq 2X_{\max}(NT)^{1/3} + 1$, so with $N\gamma = \gamma_0$ the UCB and discretisation terms are $\tilde{O}((NT)^{2/3})$. The choice of NT_0 balances $2L_f NT_0$ against the projection term $\propto d\,NT/\sqrt{NT_0}$; both become $\tilde{O}((dNT)^{2/3})$. Finally, $2L_f N_{\mathrm{b}} c_N = \tilde{O}(N(NT)^{1/3}) \leq \tilde{O}((NT)^{2/3})$ when $N \leq \sqrt{T}$. □

Remark 10 (The d-dependence: $(dNT)^{2/3}$ against DBG's $d^2T^{2/3}$)**.** For $N = 1$, Theorem 9 gives $\tilde{O}\big(d^{2/3}T^{2/3} + d^2/\mu^{*2}\big)$, which is better in d than DBG's $\tilde{O}(d^2T^{2/3})$ (their Theorem 4.3 and Corollary 4.6). The difference comes from the choice of T_0, not from a sharper estimator.

- *DBG* (Appendix C, event E_1 and Lemma C.4) require the projection error to be at most half a bin, $\|\hat{\theta}_0 - \theta^*\|_1 \leq \Delta/(2L)$. By Lemma 1 this needs $T_0 \asymp L^2 d^2 \log(d/\delta)/(\mu^*\Delta)^2 = \tilde{\Theta}(d^2T^{2/3})$ uniform rounds, each costing up to $2L_f$.
- *Here* the projection error may exceed Δ. Lemma 7 only uses that a pull into bin j has true index in the enlarged interval $\tilde{I}_j = I_j + [-\varepsilon_{\mathrm{p}}, \varepsilon_{\mathrm{p}}]$, and the enlarged intervals are allowed to overlap. The error is charged as a bias of $4L_{f'}\varepsilon_{\mathrm{p}} \propto d/\sqrt{NT_0}$ per pull, and balancing it against the $2L_f NT_0$ exploration cost gives $NT_0 \asymp (dNT)^{2/3}$.
- What remains of d^2 is the sample-size condition of Lemma 1, $NT_0 \geq (2C_\theta d/\mu^*)^2 \log(2d/\delta)$. It does not grow with T, so it is a burn-in.

Two caveats. (a) Nothing here shows $d^{2/3}$ is optimal: DBG's lower bound is for $d = 1$. (b) The leading constant is $(L_{f'} X_{\max} C_\theta/\mu^*)^{2/3} L_f^{1/3}$, so the improvement holds only once T is large enough that the burn-in is below T_0. Since it also improves the single-agent result, check it with the DBG authors before stating it as a contribution.

Remark 11 (The trigger threshold γ)**.** The factor $\sqrt{1 + N\gamma}$ is not only an artefact of the proof. With a constant γ, all N agents can each accumulate γG_J unsynced pulls in the same bin before

the trigger fires, so between syncs the network count of a bin can grow by a factor $1+N\gamma$ while each agent acts on a factor of only $1+\gamma$. That is UCB run in batches whose sizes grow geometrically with ratio $1+N\gamma$, and summing the widths over such a batch loses $\sqrt{1+N\gamma}$.

E30 (Figure 1, two links, 20 paired trials) measures this cost:

- With $\gamma = 2$ the regret relative to per-round sync grows with N: $\times 1.11$–1.12 ($N = 2$), $\times 1.32$–1.34 ($N = 4$), $\times 1.73$–1.80 ($N = 8$), $\times 2.50$–2.54 ($N = 16$). This is the trend the factor predicts.
- With $\gamma = 0.5$ the loss is at most 0.4% for every $N \le 16$, i.e. up to $N\gamma = 8$.
- The loss is *not* a function of $N\gamma$ alone. At $N\gamma = 4$ it is 11–23% for $N \in \{2, 4\}$ ($\gamma \ge 1$) but under 0.5% for $N \in \{8, 16\}$ ($\gamma \le 0.5$).

So the worst case is reached only when γ itself is large. When γ is small, the trigger fires as soon as the *first* agent reaches γG_J, and the other agents' unsynced counts are then typically well below γG_J. A bound capturing this would need an assumption on how evenly agents pull a bin; it is not attempted here. Theorem 9 needs only $N\gamma = O(1)$, so a practical choice is $\gamma = \min(0.5, \gamma_0/N)$.

The price is communication. Theorem 12 with $\gamma = \gamma_0/N$ gives $N_\mathrm{b}\big(1 + \log_{1+\gamma_0/N} NT\big) \approx N_\mathrm{b}\, N \ln(NT)/\gamma_0$ sync rounds, linear in N. The experiments are consistent with the theorem: at a single N a constant γ is the case $\gamma_0 = N\gamma$. E22 ($\gamma = 0.5$, $N \le 16$) has $\gamma_0 \le 8$.

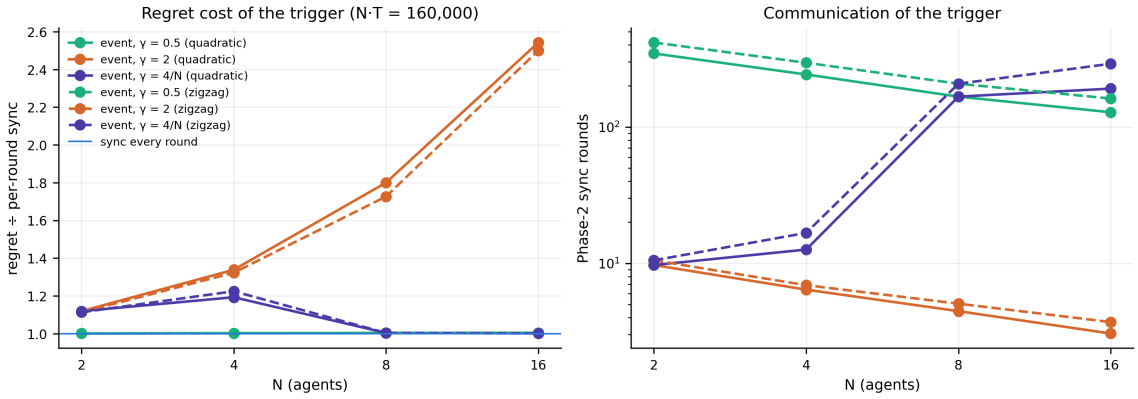


Figure 1: Constant γ against $\gamma = \gamma_0/N$ (E30, $NT = 160\,000$, 20 paired trials, $\gamma_0 = 4$). Left: regret relative to per-round sync; $\gamma = 2$ degrades with N, $\gamma = 0.5$ does not. Right: Phase-2 sync rounds; all event triggers use 3–417 against 9 940–79 520 for per-round sync.

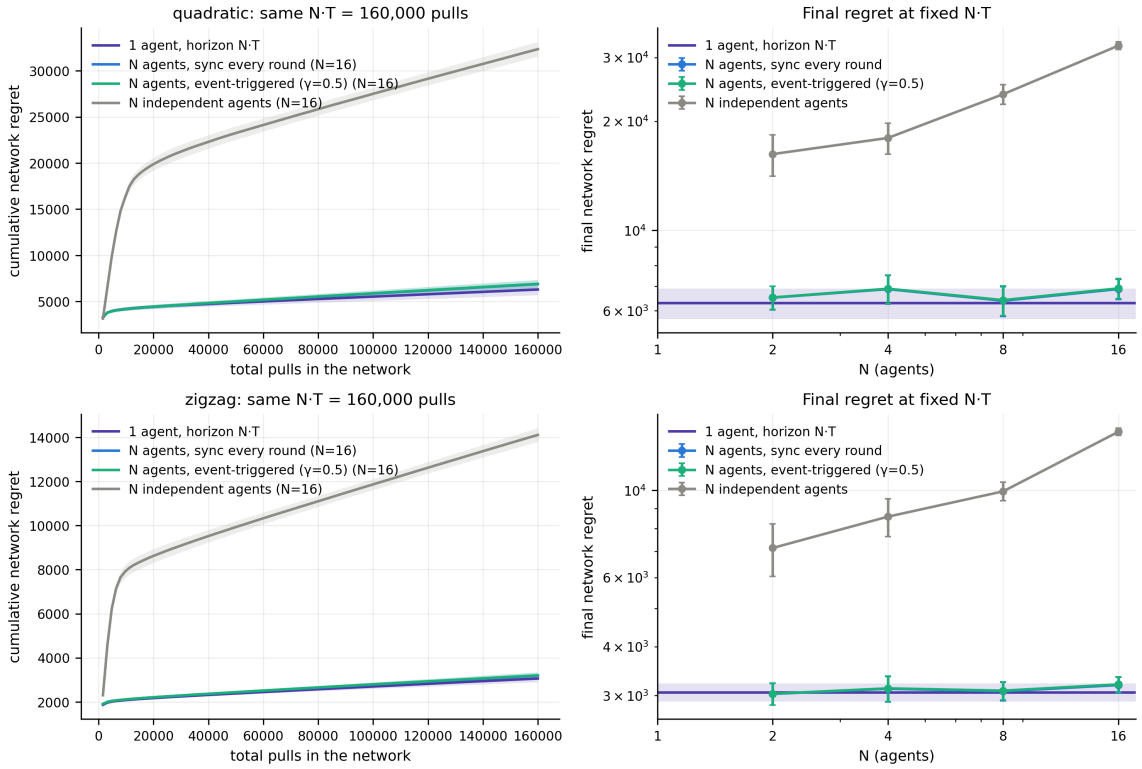


Figure 2: Empirical check of Theorems 3 and 9 (E22, 20 paired trials). Fixed total budget $NT = 160\,000$, same pooled exploration and bin width for every N. Left: regret against total pulls; the single agent at horizon NT and $N = 16$ synced agents overlap. Right: final regret against N; synced agents stay within 0–10% of the single agent and never beat it, while independent agents are 2.3–5.1× worse.

4 Communication

Theorem 12 (Sync rounds)**.** *Phase 2 of Fed-ZoomSIB-SS performs at most*

$$\sum_{j \text{ used}} \left(1 + \log_{1+\gamma} n_j\right) \;\le\; N_{\mathrm{b}}\left(1 + \log_{1+\gamma} \frac{NT}{N_{\mathrm{b}}}\right)$$

sync rounds, where n_j is the final number of pulls in bin j. Per-round sync uses $T - T_0$.

Proof. Charge each sync to one bin j whose trigger fired, i.e. $\delta n_{l,j} > \gamma \max(G_j, 1)$ for some l. The sync adds at least $\delta n_{l,j}$ to G_j:

- if $G_j = 0$, it becomes ≥ 1;
- otherwise G_j grows by a factor of at least $1 + \gamma$.

Since $G_j \le n_j$, bin j is charged at most $1 + \log_{1+\gamma} n_j$ times. Bins never pulled are never charged. Concavity of log and $\sum_j n_j \le NT$ give the second form. □

Corollary 13 (Scalars)**.** *Each sync sends 3 scalars per (agent, touched bin) up and 3 per merged bin to each of N agents down. Every pull is uploaded once, so uploads total at most $3NT$. Downloads total at most $3NN_{\mathrm{b}}$ per sync, so at most $3NN_{\mathrm{b}}^2(1 + \log_{1+\gamma} NT)$ overall, which is $\tilde{O}(N(NT)^{2/3})$ for $\Delta = (NT)^{-1/3}$. Per-round sync downloads up to $3N \min(N, N_{\mathrm{b}})(T - T_0)$.*

Remark 14 (The bound is loose in practice)**.** E22 measures 61–213 sync rounds for $NT = 10^4$–$3.2 \cdot 10^5$ ($N = 8$), against a bound of 3 600–15 600, i.e. 60–75× lower. The busiest bin triggers about 3 syncs, against $1 + \log_{1+\gamma}(NT) \approx 24$–$32$. The worst case of Theorem 12 assumes every bin is pulled heavily. Sleeping UCB concentrates pulls on a few near-optimal bins, and most bins trigger at most once. A gap-dependent communication bound is open (Section 6).

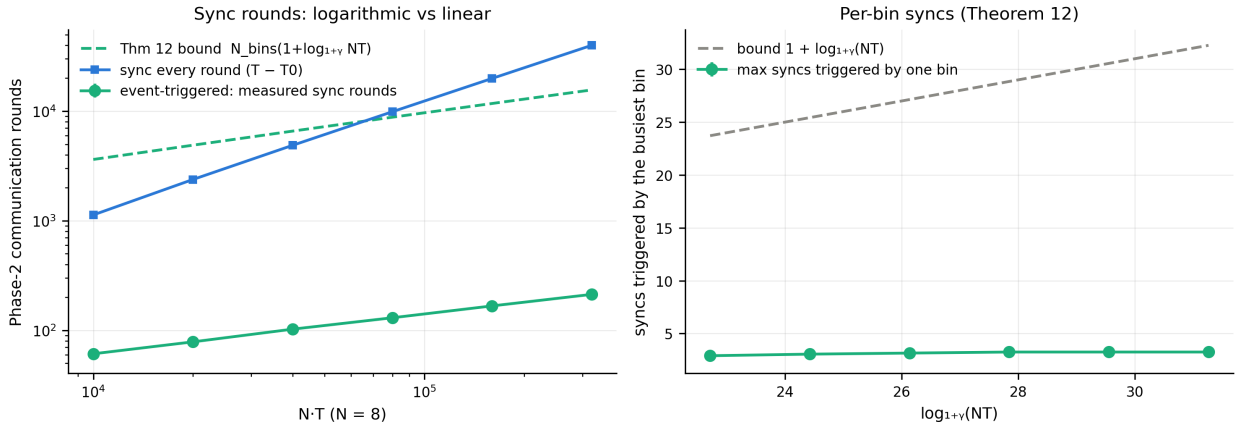


Figure 3: Empirical check of Theorem 12 (E22 part 2, $N = 8$). Left: Phase-2 sync rounds of the event trigger grow logarithmically (61→213 as NT grows 32×) while per-round sync grows linearly; the theorem's bound (dashed) is 60–75× above the measurement. Right: the busiest bin triggers about 3 syncs against a per-bin bound of 24–32.

Proposition 15 (Phase 1 is a vanishing share)**.** *With fixed T_0, Phase 1 sends $N(2d+3)$ scalars: one $(d+1)$-vector up per agent,* $\max|z|$ *up, and $(\hat{\theta}_0, W)$ down. Suppose $\gamma < 1$, and let $N_{\mathrm{b}}^{\mathrm{used}}$ be the number of bins ever pulled in Phase 2. Then Phase 2 sends at least $3NN_{\mathrm{b}}^{\mathrm{used}}$ scalars, so*

$$\frac{\textit{Phase-1 scalars}}{\textit{all scalars}} \leq \frac{2d+3}{2d+3+3N_{\mathrm{b}}^{\mathrm{used}}}.$$

Proof. With $\gamma < 1$, a first pull into an empty bin makes $\delta n = 1 > \gamma \max(0, 1)$, so the bin is merged at the next sync and broadcast (3 scalars) to all N agents at least once. □

Since $N_{\mathrm{b}}^{\mathrm{used}}$ grows like $(NT)^{1/3}$ until most of the window is visited, the share is $O(d/(NT)^{1/3})$. Phase 1 matters only when d is comparable with $(NT)^{1/3}$. E24 measures under 2% at $d \leq 100$, and compressing Phase 1 saves at most that share.

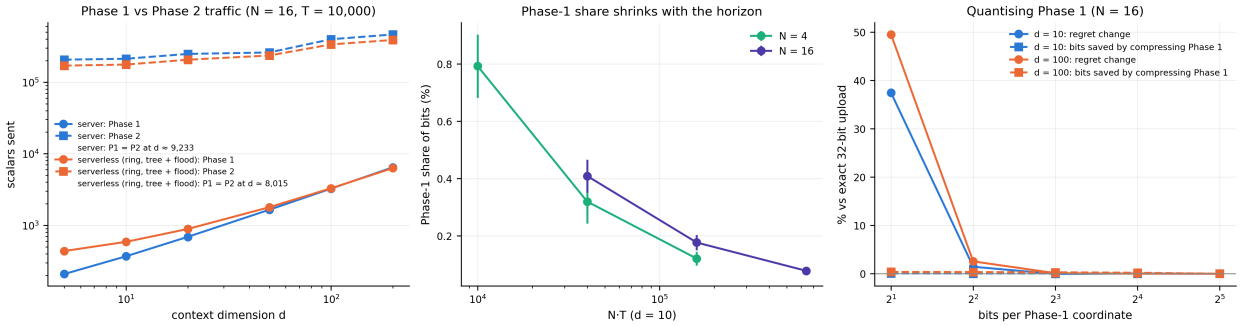


Figure 4: Empirical check of Proposition 15 (E24). Left: Phase-1 traffic grows linearly in d, Phase-2 traffic is far larger for every $d \le 200$, server and serverless. Middle: the Phase-1 share falls as NT grows, as $O(d/(NT)^{1/3})$ predicts. Right: quantising the Phase-1 upload saves at most 0.06–0.38% of all bits, while 2-bit rounding costs 37–50% regret.

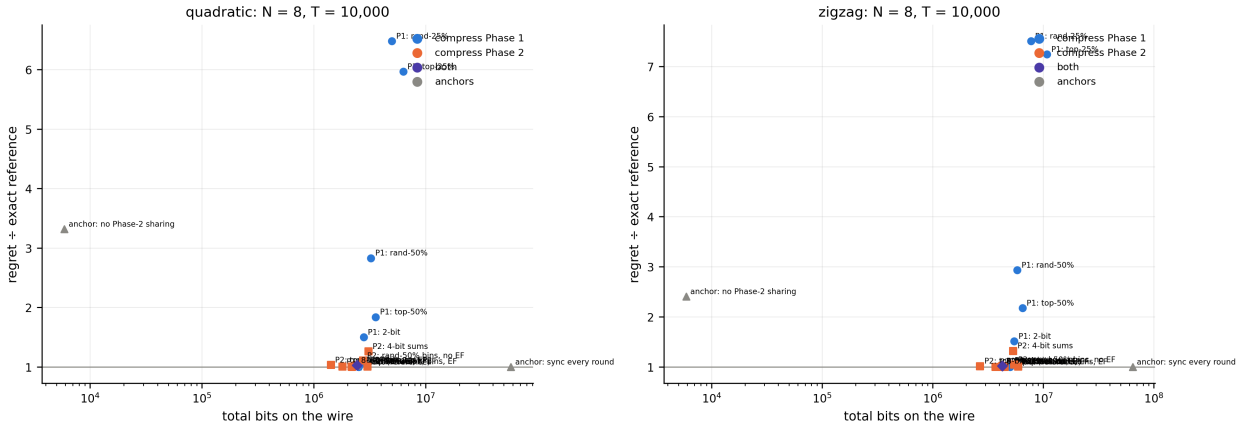


Figure 5: Compression on each phase (E26, $N = 8$, 20 paired trials): regret relative to the exact protocol against total bits on the wire. Compressing Phase 1 (circles) never helps: a noisier frozen direction makes Phase 2 sync more. The only free saving is sending fewer Phase-2 bin messages (top-25% bins with error feedback: 26–29% fewer bits).

5 Without a server: the relay version (sketch)

Dec-ZoomSIB-Relay replaces the server as follows:

- spanning-tree aggregation in Phase 1, which is exact, so Lemma 5 holds after 2·depth extra exploration rounds;
- event-triggered *flooding* of each agent's own bin table in Phase 2: records travel one hop per round, and agent l republishes when its unpublished count exceeds γ times its current view.

Theorem 16 (Relay, sketch)**.** *On a connected graph of diameter D, with the parameters of Theorem 9, the relay version satisfies* $\mathbb{E}R_T(N) \le$ (*bound of Thm. 9*)$+O(L_f\, ND\, N_b)$, *i.e.* $\tilde{O}\big(d^{2/3}(NT)^{2/3} + ND(NT)^{1/3}\big)$.

Proof sketch. Lemma 8 gains one more kind of unseen pull: records published by agent l fewer than $\mathrm{dist}(i,l) \le D$ rounds ago. Each agent makes at most one pull per round, so these number at most

$(N-1)D$ per bin. Then $n_{i,J} \geq (k - c_N - ND)/(1 + N\gamma)$, and the first $c_N + ND$ pulls of each bin are charged $2L_f$. Lemma 6 needs the union bound to run over each agent's *view*, which is a union of N per-agent prefixes; this costs a factor $\sqrt{N}$ inside the log-width argument. □

Open. Whether the additive NDN_b is necessary is unknown. E23 ($N = 16$) measures an excess of +1.5% ($D = 1$) to +9% ($D = 15$) that grows sublinearly in D, is almost flat in T (log–log slope 0.13–0.14 against 1/3), and is 60–150× below NDN_b. This suggests the bound is loose: plausibly only Phase-1 delays and the first visits to each bin pay the diameter.

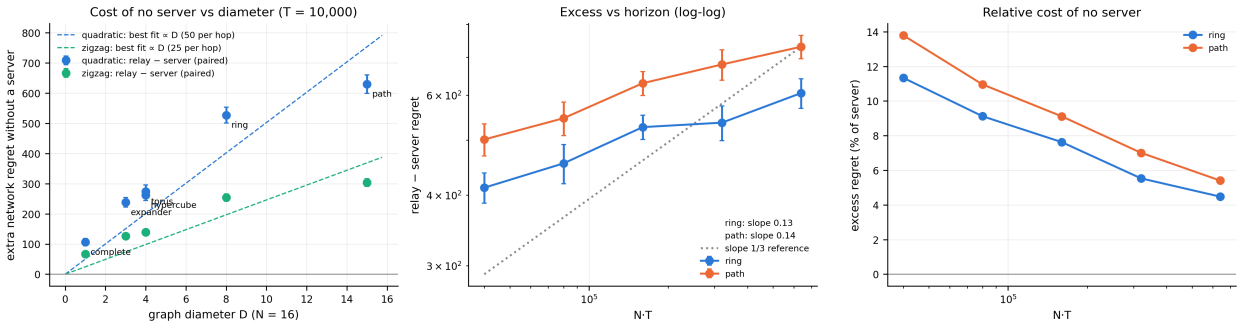


Figure 6: Theorem 16 against measurement (E23, $N = 16$, 20 paired trials). Left: extra regret without a server grows sublinearly in the diameter D (dashed: best fit $\propto D$). Middle: the excess is almost flat in NT (slope 0.13–0.14 against the 1/3 the bound predicts). Right: as a fraction of the server's regret it falls with the horizon.

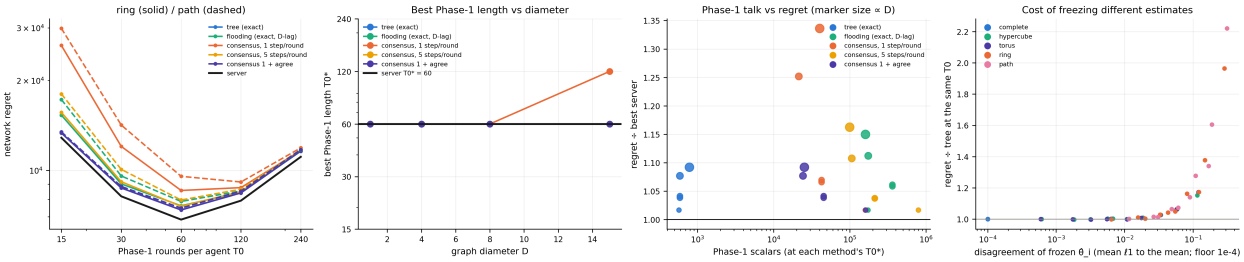


Figure 7: Exploration length and agreement on graphs (E28, $N = 16$, 10 paired trials). From left: regret against the Phase-1 length T_0 on the ring and path; the best T_0 barely moves with D; Phase-1 communication against regret; and the regret cost of agents freezing different estimates, which one agreement pass at the freeze removes.

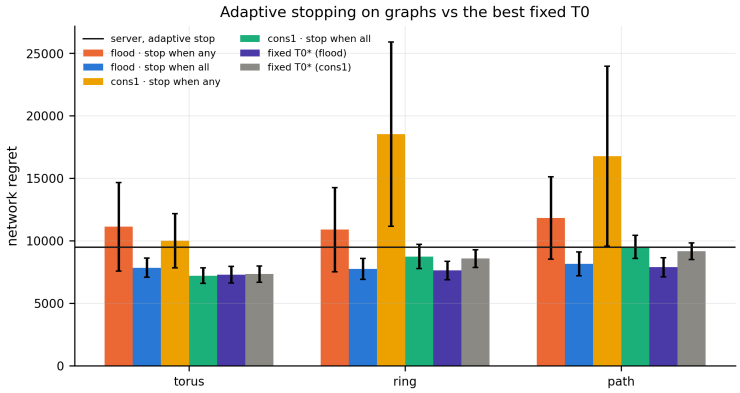


Figure 8: Adaptive stopping on graphs (E28). "Stop when any agent is stable" costs 1.4–2.2× the regret of a fixed T_0; "stop when all are" is close to it but is modelled without its $\geq D$-round detection delay.

6 The regret–communication trade-off (notes and conjecture)

Theorems 9 and 12 give *one* point on the trade-off curve: optimal regret with $O(N_{\mathrm{b}} \log NT)$ sync rounds. The natural question is the whole curve: with a budget of C sync rounds, what is the best achievable regret?

A heuristic lower bound. Between two syncs, agents share no information. Consider a sync-free window of length L that starts after the network has pooled n samples. During it, the N agents behave like N learners with a common warm start, each learning only from its own L pulls. In the single-agent hard instance (Lemma 2), the bins that must be distinguished have gaps of order $(n+L)^{-1/3}$, and a lone learner needs $L^{2/3}$-type regret to resolve them. That suggests

$$R_T(N;C) \;\gtrsim\; \max\Big\{(NT)^{2/3},\; \max_{\text{windows}} N\,L^{2/3}\Big\},$$

where the largest window is at least T/C if syncs are spread evenly. Under this heuristic, evenly spread syncs need $C \gtrsim \sqrt{N}$ to keep the second term below the first. Windows that grow geometrically could need far fewer.

Conjecture 17. For single-index bandits with a server, the optimal regret $\tilde{\Theta}((NT)^{2/3})$ is achievable with polylog(NT) sync rounds independent of N_{b}. For batched Lipschitz bandits, $O(\log\log T)$ batches suffice (Feng, Huang & Wang 2022; to be checked), but that algorithm eliminates arms it chooses to play. Here arms are given by contexts (sleeping bins), which is exactly what our fixed-schedule experiments (E27) find breaks naive batching.

What a paper would need: (a) a batch-aware Phase 2 (e.g. elimination over bins with context-driven availability) with an upper bound in terms of C; (b) a matching lower bound that formalises the window heuristic. Neither is done here.

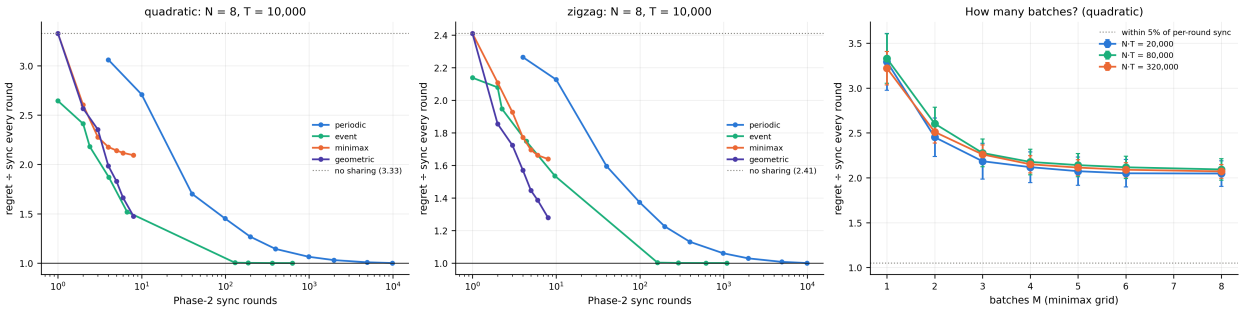


Figure 9: The regret–communication frontier (E27, $N = 8$, 20 paired trials). Left and middle: regret relative to per-round sync against the number of Phase-2 sync rounds for periodic, event-triggered and fixed batched schedules. The event trigger reaches $\times 1.004$ with about 130 syncs; periodic sync with 99 syncs is at $\times 1.37$–1.45. Right: with the minimax (Perchet) grid, 8 batches still leave $\times 2.1$ at every horizon, so $O(\log \log NT)$ batches do not suffice for these agents.

Where the experiments are

- Theorem 3/9 in practice: E22 (Figure 2). One agent at NT against N agents at T with the same pooled exploration and bin width.
- Theorem 12: E22 part 2 (Figure 3), sync rounds and per-bin triggers against $\log NT$.
- Remark 11 (trigger threshold): E30 (Figure 1).
- Proposition 15: E24 and E26 (Figures 4 and 5).
- Theorem 16: E23 and E28 (Figures 6, 7, 8).
- Section 6: E27 (Figure 9).